COMPOUND 1: NITRATES (NO_3^-)

Justification :- Detected in Martian sediments by Curiosity rover. It indicates a source of Nitrogen (element of life). This suggests potential extinct or extant microbial life.

Test :- Nitrate Detection using Diphenylamine.

Test Reaction: Diphenylamine + Nitrate (in Sulphuric Acid) → Blue Complex

Pros: very sensitive and gives very fast results (~2 – 3 minutes).

Cons: Requires conc. H_2SO_4 (corrosive).

Rover viability: Use preloaded gel cartridges with encapsulated acid. The test should be sealed in a reaction chamber.

Reference :- https://extension.missouri.edu/publications/g9811

COMPOUND 2: ORGANIC ACIDS (ex: Acetic acid)

Justification :- Organic Molecules have been previously found on Mars in sediment samples which can represent microbial metabolism. Again, it might point to some extinct or extant life on Mars.

Test :- Sodium Bicarbonate Test.

Test Reaction: Add soil to dil. $NaHCO_3$ → CO_2 effervescence if acid is present.

Pros: No toxic reagents and easy visual analysis due to bubbling.

Cons: It is not specific to any type of acid.

Rover viability: Combine soil and $NaHCO_3$ in transparent chamber, use camera to monitor fizzing / effervescence.

Reference :- https://theory.labster.com/sodiumbi_test/

COMPOUND 3: PHOSPHATES (PO_4^{3-})

Justification :- Phosphorous was detected by Phoenix lander. It is important for analysis as Phosphorus is crucial for DNA/RNA structure. Its presence indicates potential for life-supporting chemistry.

Test :- Phosphate Detection using Ammonium Molybdate.

Test Reaction: Phosphate + $(NH_4)_2MoO_4$ (in Nitric Acid) → Yellow Precipitate

Pros: Easily studied due to visible change.

Cons: Needs acid reagent.

Rover viability: Reaction in microfluidic chip with controlled acid volume. Detect yellow precipitate using optical sensor.

Reference :- https://pubs.geoscienceworld.org/sepm/jsedres/article-abstract/20/2/116/95071/A-spot-test-for-phosphorus-in-rocks?redirectedFrom=PDF